

# LLM Encapsulation in Sage

# Overview

- Background Information
- Coding
  - My experience in setting up and deploy an LLM on the Sage Node
  - Going over, what I assume to be, the important aspects
- ECR Deployment
  - Quick Start for Deploying Moondream 2 on ECR

# Background Information

- What is Moondream
  - Moondream is a tiny vision language model typically meant for low resource availability devices like edge devices
  - Inputs typically are image + prompt and output can be a natural language response, point detection, and object detection
  - <https://moondream.ai/>
- What is Moondream in Sage?
  - Moondream in Sage is basically having taken the moondream model and wrapping it in a container that is deployable on Sage Nodes.
  - <https://github.com/cleeuh/moondream-for-sage-node>

# LLM Encapsulation in Sage Coding

# Dockerfile

## Dockerfile

```
# Technically moondream's python client now uses Onnxruntime for inference, so torch is not needed...
FROM waggle/plugin-base:1.1.1-ml-torch1.9

WORKDIR /app

ENV DEBIAN_FRONTEND=noninteractive
ENV PYTHONUNBUFFERED=1

# Initial Deps
RUN apt-get update --fix-missing
RUN apt-get install -y wget curl pkg-config libssl-dev curl

# Update GCC via REPO (perhaps you can use this hacky way to bypass the need to compile the aforementioned)
RUN echo "deb http://ports.ubuntu.com/ubuntu-ports jammy main" | tee /etc/apt/sources.list.d/temporary-repository.list

# Python 3.10
RUN apt-get update --fix-missing
RUN apt-get install -y build-essential python3 python3-pip

RUN curl -sS https://bootstrap.pypa.io/get-pip.py | python3
RUN python3 -m pip install --upgrade pip

# Python Dependencies
COPY requirements.txt .
RUN pip3 install --no-cache-dir -r requirements.txt
RUN wget https://github.com/cleeh/moondream-for-sage-node/releases/download/resources/onnxruntime_gpu-1.19.0-cp310-cp310
RUN pip3 install onnxruntime_gpu-1.19.0-cp310-cp310-linux_aarch64.whl
RUN pip3 install numpy==1.26.4

# Model
RUN wget https://github.com/cleeh/moondream-for-sage-node/releases/download/resources/moondream-2b-int8.mf -O moondream-
RUN wget https://github.com/cleeh/moondream-for-sage-node/releases/download/resources/moondream-2b-int4.mf -O moondream-
RUN wget https://github.com/cleeh/moondream-for-sage-node/releases/download/resources/moondream-0_5b-int8.mf -O moondream-
RUN wget https://github.com/cleeh/moondream-for-sage-node/releases/download/resources/moondream-0_5b-int4.mf -O moondream-

COPY . .

# Space Reduction
RUN rm -rf /var/lib/apt/lists/*

# # ENTRYPOINT [ "/usr/bin/bash" ]
ENTRYPOINT ["python3", "main.py"]
```

- Base Image: waggle/plugin-base:1.1.1-ml-torch1.9
- Updating to newer python version by referencing newer repository or compiling.
  - For less headaches, I recommend trying to either reference a newer repo or installing from local (.deb files) before trying to compile from source.
  - Newer python versions will support more up to date packages (pytorch 2.x, onnxruntime, etc.)
- Be cognizant of ARM vs x86 repos

# Arguments

main.py

```
def parse_args():
    parser = argparse.ArgumentParser(description='Moondream 2B in
    parser.add_argument('--stream', type=str, default="bottom_camera")
    parser.add_argument('--continuous', action='store_true', default=False)
    parser.add_argument('--caption', action='store_true', default=False)
    parser.add_argument('--dynamic-loading', action='store_true', default=False)
    parser.add_argument('--small', action='store_true', default=False)
    parser.add_argument('--int4', action='store_true', default=False)
    parser.add_argument(
        '--query',
        action='append',
        default=[],
        help='Prompt the model and get a response'
    )
    parser.add_argument(
        '--detect',
        action='append',
        default=[],
        help='Bounding boxes for the image from a prompt'
    )
    parser.add_argument(
        '--point',
        action='append',
        default=[],
        help='Get X,Y location for the image from the prompt'
    )
    return parser.parse_args()
```

- Use argparse library
  - `import argparse`
- Make sure arguments are also noted in sage.yaml

sage.yaml

```
name: "moondream-2"
version: "0.2.0"
description: "Moondream 2b with CPU inferencing for Sage nodes"
keywords: "llm"
authors: "Christopher Lee, Jason Leigh"
collaborators: "Laboratory for Advanced Visualization and Applications"
funding: "RAPID Award NSF 2346568"
license: "MIT"
homepage: "https://github.com/cleeuh/moondream-for-sage-node"
source:
  architectures:
    - "linux/arm64"
inputs:
  - id: "stream"
    type: "string"
    description: "ID or name of a stream, e.g. bottom_camera, top_camera"
  - id: "continuous"
    type: "boolean"
    description: "Flag to run this plugin forever"
  - id: "caption"
    type: "boolean"
    description: "Generate a caption from the model"
  - id: "small"
    type: "boolean"
    description: "Load the 0.5b model instead of 2b"
  - id: "int4"
    type: "boolean"
    description: "Use quantized int4 instead of int8"
  - id: "dynamic-loading"
    type: "boolean"
    description: "Load and unload parts of the model as needed"
  - id: "query"
    type: "string"
    description: "Prompt the model and get a response"
  - id: "detect"
    type: "string"
    description: "Bounding boxes for the image from a prompt"
```

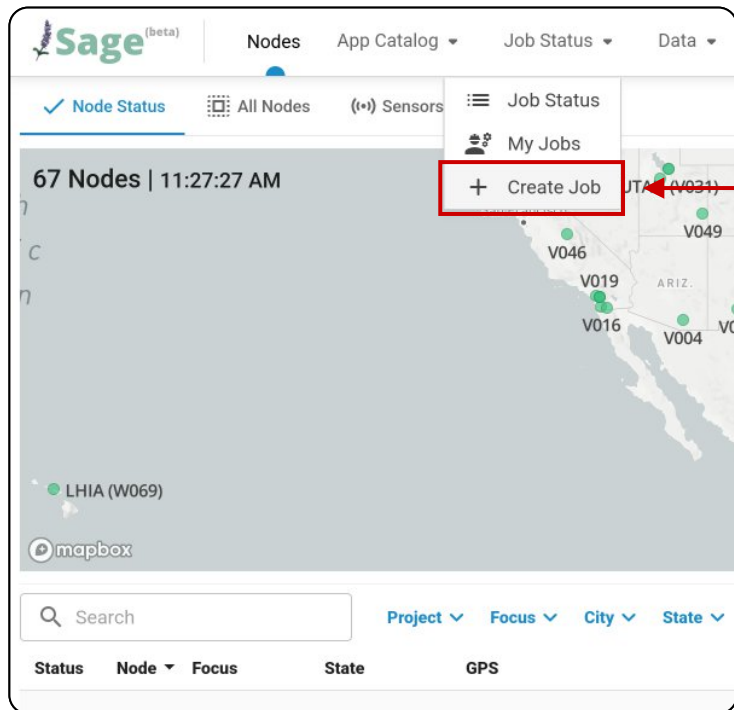
# Additional Remarks

- When build on local system, make sure to build and test for the arm platform
  - <https://docs.docker.com/build/building/multi-platform/>
- There maybe compatibility issues where GPU remains unutilized
  - Use resource selector gpu
    - `sudo pluginctl run --name plugin-objectcounter --resource request.cpu=1,limit.cpu=2,request.memory=2Gi,limit.memory=3Gi --selector resource.gpu=true 10.31.81.1:5000/local/plugin-objectcounter -- -stream bottom_camera -all-objects`
  - Outdated firmware on older nodes
    - newer orin nodes with latest jetpack should streamline the process

# LLM Encapsulation in Sage ECR Deployment



# Quick Start for Deploying Moondream on ECR



- Go to: <https://portal.sagecontinuum.org>
- Make sure you are logged in
- Click [Job Status] then [Create Job]

## My jobs / Create job (Science Goal)

{ } Editor

```

1 name: moondream-test-job
2 plugins:
3   - name: moondream-test-app
4     pluginSpec:
5       image: registry.sagecontinuum.org/chrislee/moondream-2:0.2.0
6     args:
7       - --small
8       - --stream
9       - bottom_camera
10      - --query
11      - Is there a fire?
12     selector:
13       resource.gpu: "true"
14     resource:
15       limit.cpu: "1"
16       limit.memory: 2.9Gi
17       request.cpu: "1"
18       request.memory: 2.9Gi
19     volume: {}
20 nodeTags: []
21 nodes:
22   W070: null
23 scienceRules:
24   - 'schedule(moondream-test-app): cronjob("moondream-test-app", "**/30 * *
25   successCriteria: []
26

```

Node to run on

## Arguments (Args):

description: ID or name of a stream, e.g. bottom\_camera, top\_camera, left\_camera

id: stream

type: string

description: Flag to run this plugin forever

id: continuous

type: boolean

description: Generate a caption from the model

id: caption

type: boolean

description: Load the 0.5b model instead of 2b

id: small

type: boolean

description: Use quantized int4 instead of int8

id: int4

type: boolean

description: Load and unload parts of the model as needed

id: dynamic-loading

type: boolean

description: Prompt the model and get a response

id: query

type: string

description: Bounding boxes for the image from a prompt

id: detect

type: string

description: Get X,Y location for the image from the prompt

id: point

type: string

Submit Job

Submit when done

- You can verify that it is running in [Job Status] -> [My Jobs]
- Note that smaller models, tend to hallucinate more when prompted.

Apps

Names

Images

Audio

App Filters

Apps: \*moondream-2:0...

Nodes: W070

Names

cURL download

curl https://data.sagecoi

Python snippet

```
import sage_data_client
df = sage_data_client.query(
    start="-7d",
    ...
)
```

Clear apps is \*moondream Quick Ranges last 7 days 10/29/2025 8:59:44 AM to 11/5/2025 8:59:44 AM

relative time meta Filter Values 1-100 of 738

| Node | Time         | Name   | Value                                                                                                    |
|------|--------------|--------|----------------------------------------------------------------------------------------------------------|
| W070 | 29.1 min ago | query  | [{"query": "Is there a fire?", "response": "Yes, there is a fire burning in the middle of the forest."}] |
| W070 | 29.1 min ago | upload |                                                                                                          |

## W070

Birch Hill, 21941 Crestline Rd, Palomar Mountain, CA 92060



# Examples

```
name: moondream-test-job
plugins:
- name: moondream-test-app
  pluginSpec:
    image: registry.sagecontinuum.org/chrislee/moondream-2:0.2.0
    args:
      - --small
      - --stream
      - bottom_camera
      - --query
      - Is there a fire?
  selector:
    resource.gpu: "true"
  resource:
    limit.cpu: "1"
    limit.memory: 2.9Gi
    request.cpu: "1"
    request.memory: 2.9Gi
  volume: {}
nodeTags: []
nodes:
  W070: null
scienceRules:
- 'schedule(moondream-test-app): cronjob("moondream-test-app", "*/30 * * * *")'
successCriteria: []
```

```
name: moondream-test-job
plugins:
- name: moondream-test-app
  pluginSpec:
    image: registry.sagecontinuum.org/chrislee/moondream-2:0.1.5
    args:
      - --dynamic-loading
      - --stream
      - bottom_camera
      - --caption
      - --point
      - smoke
      - --point
      - fire
      - --point
      - hazard
      - --query
      - Is there a fire?
  selector:
    resource.gpu: "true"
  resource:
    limit.cpu: "3"
    limit.memory: 7Gi
    request.cpu: "3"
    request.memory: 7Gi
  volume: {}
nodeTags: []
nodes:
  W070: null
scienceRules:
- 'schedule(moondream-test-app): cronjob("moondream-test-app", "*/30 * * * *")'
successCriteria: []
```

# Other Notes

- Job may not run immediately after submission as the node must download the container.
- Enabling the GPU in the resource selector as the code doesn't work reliably on Raspberry Pi Compute Units.
  - Moondream 2, as of now runs on CPU only
- Dynamic Loading is meant to reduce the RAM usage slightly with trade off for longer compute time.
- Moondream 2 model used is the 2024 variant
- Smaller model constraints
  - Point and Object Detect wont work
- Sophisticated prompts may not work reliably, keep prompts short and concise for best results